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Windows CE is Ready, But for What?

The embedded market has been traditionally commandeered by those producing tight code, integration, compact real estate, and low cost. However, Microsoft wants to enter the picture. Edward questions whether they can adjust.

Microsoft's launch of the Windows CE operating system for embedded systems raises the question, "How does Microsoft define the embedded-systems market?"

True embedded PCs are generally characterized as single-purpose computers. Even if end users know a PC-compatible computer is embedded in the device, they don't interact with the device as a desktop computer.

However, Microsoft only plans to build Windows CE into devices which are merely smaller versions of desktop PCs (e.g., palm-tops and laptops).

As well, in the traditional real-time embedded computing market, most electronic-equipment manufacturers make products in the range of 300–3000 per year. Microsoft sells direct to companies whose volume is 25,000 units a year or greater. Anything less goes through a distributor [1].

Only in consumer products do we see such high levels of usage. Is Microsoft ready to take on the additional support and lower volume that are inevitable parts of the embedded-PC market?

WINCE AS AN EMBEDDED KERNEL

The announcement of Windows CE as a platform for real-time embedded applica-

tions has both good and bad aspects.

The good part is the publicity Win32 will receive. Phar Lap Software has been supporting real-time Win32, first in the TNT DOS-Extender and then in the ETS Kernel. It was, in fact, the first to support the Win32 API native with an embedded kernel.

Today, QNX, Wind River, and other vendors have added a Win32 API layer to their embedded kernels—the APIAccess Win32 layer (developed by Willows, but now owned by Award Software). And, VenturCom is shipping a real-time kernel using the Win32 API as a layer for Windows NT.

Support is growing for real-time embedded applications development using Visual C++ and Borland C++ compilers. Windows CE will accelerate that movement. For years, using common compilers and a single API has been a dream in embedded computing.

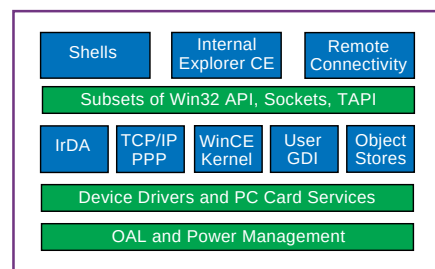


Figure 1—Any of the elements in the Windows CE architecture can be replaced or removed for embedded systems.

On the down side is Windows CE's lack of "embeddedness" and true real-time functionality [2]. Windows CE suits small portable computer products, but it falls apart as a blood analyzer, robot, machine tool, or other intelligent machine.

Its strength lies in its WinNT/95 look and feel. But as an embedded kernel, it lacks the pluses of real-time kernels like Wind River's VxWorks or Phar Lap's Real-time ETS Kernel.

Compared to Windows NT, WinCE is small—less than 1 MB. A typical kernel should use 500-KB ROM and 100-KB RAM [1]. However, it's also a multiprocess system with the inherited overhead of Windows 95.

Microsoft is targeting portable computing and automotive applications—places where the product being sold is a computer. Given the Windows look and feel, ease of development, and general acceptance of Windows, it should do well in these markets.

MISSING FEATURES

For WinCE, Microsoft supports a subset of the Win32 API (about 500 functions). No new APIs were added, according to R. Kanemori, Microsoft Senior OEM Account Manager. By contrast, Phar Lap Software uses 140 of the Win32 APIs in its product and felt the API set needed another 140 functions for embedded and real-time applications.

Without added APIs, how does Microsoft expect you to initialize a chipset if there's no BIOS? How can you set an event flag in an ISR, or track the thread-execution sequence in a complex system? How do you change the system-clock frequency or time slice?

With Windows CE, you can't do any of these using a high-level language and ready-made functions. Both the Win32 API and Windows CE kernel need these additions to properly support real-time and embedded applications.

DEVELOPMENT TOOLS

Development requires Visual C++ for Windows CE—a special cross-development version of Visual C++. In addition, OEMs must purchase the OEM Adaptation Kit (OAK) containing the Windows CE code.

The OAK is free to OEMs when they purchase a sufficient quantity of CE kernel licenses. According to one customer, cost is \$70 per 1000. It's intended to let the OEM write or modify the OEM Abstraction Layer

(OAL) built between the hardware and the rest of the OS (sound like Windows NT?).

According to Annabooks, a Windows CE kernel distributor, you also need Microsoft's device-driver development kit (DDK) [3]. Eclipse, another CE distributor, recommends their Eclipse development system for debugging and execution [4].

To emulate the target environment during development, you have to develop on a Windows NT system. A set of special development DLLs is provided with Visual C++ for Windows CE for this purpose.

The target requires built-in networking to debug via a serial line supporting UDP or TCP/IP PPP. According to Microsoft, the UDP transfer method isn't always reliable. The chip or board vendor will probably have to provide this support. Today (prior to 2.0), TCP/IP has no general network support.

OTHER PROBLEMS

And, there are other drawbacks. For one, the system supports Unicode only.

Unicode is the 16-bit representation of characters that stores and displays Latin and Asian character sets. This feature is important for local language displays. But, most intelligent machines have minimal displays that don't require such a feature.

For the OEM, Unicode has other problems besides its double memory space. If an application needs to parse a string or search a text array, the software must be rewritten to accommodate this wide-character storage. It will be difficult to rewrite existing C applications to accommodate Unicode.

Priority scheduling in Windows CE is different from both Windows NT and Windows 95. Unlike Windows NT's four priority classes, there's just one class of priorities in the CE kernel and only eight levels (0–7). Microsoft indicates that there are only seven levels (0–7, minus level 6) [5].

Windows CE V.1.01 is out now. In the second quarter of 1997, V.2.0 was scheduled for beta testing, with final versions shipping by the end of September 1997.

The old hat that Microsoft V.3.0 is the first useful version may hold true here as well. Even Microsoft says, "...the 2.0 version may be more appropriate for some embedded systems designers than 1.01" [6]. V.2.0 will be the first to support the '486 and PowerPC.

WHAT'S MISSING?

The Windows CE system supports a subset of the Win32 API. Nothing new has

been added. And, many useful APIs or functions are lacking.

First of all, there's no way to initialize the CPU and support chips other than the few odd chips currently supported (i.e., the chip manufacturer or OEM must write their own BIOS). Developers of custom boards using chips other than the reference designs have no easy way to initialize their chipset.

You can't set events or semaphores from an ISR. This feature would allow for short and fast ISRs, permitting the kernel to start threads waiting for events produced by ISRs.

There's no API to suspend the scheduler, so the application can't raise itself above the kernel to execute a critical function without the possibility of the kernel resuming and passing control to another thread. But, without the ability to set events and semaphores from ISRs, this is of less importance.

In both Windows 95 and NT, the kernel can take control over even the highest-priority thread—not a feature you want in a critical application. VenturCom plans to include scheduler suspension in its real-time Windows CE kernel.

Windows CE also doesn't include an API for priority-inversion avoidance (a quirk resolved in every serious real-time kernel). Will Microsoft solve this problem for WinCE like it did in both Windows NT and Windows 95 kernels [7]?

The Windows 95 kernel senses when a higher priority thread is blocked from access to a critical section by a lower priority thread blocked from running by a mid-level priority thread. The kernel temporarily raises the priority of the lower priority thread to let it complete its operation and release the critical section. VenturCom plans to have priority-inheritance avoidance in its version of the Windows CE kernel.

Changing the system clock's frequency is also not supported. This ability lets the kernel have control more often, and increasing the frequency enables finer control of functions.

There also isn't an API for changing the length of the round-robin time slice. In both Windows NT and Windows 95 kernels, threads executing at the same priority enter into a round-robin scheduler.

This setup gives each thread CPU execution time. But since I can't find documentation for a round-robin scheduler, I doubt there's a way or need to change the time slice.

WinCE does not support sequence-of-events logging, either, which offers a se-

quence list of thread execution and the parameters they pass as the kernel activates them or they call kernel functions.

This debugging tool is useful when many threads have different priorities. A sequence-of-events logging mechanism is the only way to find the sequence that a set of threads actually executed.

Direct access to I/O registers requires kernel-level drivers to support new or different hardware. Hence, there are lengthy context switch times when the application accesses I/O.

Microsoft says thread-switch times are less than 100 μ s (about four times slower) than most real-time embedded kernels. VxWorks and others use a single process (single address space, everyone resides in ring 0) system, making context switches easy and fast.

Application-level drivers don't require a rigid driver format. Windows CE has a registry system similar to Windows 95, where applications can be registered with the kernel and receive interrupts directly, but that's not the same as getting maximum throughput via direct access to I/O registers.

As well, WinCE does not support a full-function embedded Web server. There isn't enough RAM or ROM for complete TCP/IP support and a Web server, but networking is expected to be available in V.2.0.

The only announced Web server for the product has no local file access, so it can't send HTML pages stored in the system. The kernel needs to support in-memory files to emulate a Web server's function. (See Phar Lap's "world's smallest Web server.")

The universal GUI—the Web browser—is used in standard CE systems to communicate as Web PCs with intelligent embedded systems. Maybe Spyglass's WWWAccess

will rescue Microsoft here, too.

However, because of the modular architecture of Windows CE (see Figure 1), most of these missing functions can be added or existing functions modified.

Windows CE has so much overhead that Microsoft won't support its kernel on '386 CPUs. But, the Intel '386 EX and RadiSys EXPLR2 board are popular for developers.

By the end of September, we should have seen support for generic '486s (if virtual memory is supported), Pentium, and PowerPC chips. But, the only chips supported now are the Hitachi SH-3, NEC Vr4100 series, and Philips PR31500 (MIPS R3000 RISC core) and UCB1100—not exactly mainstream with embedded-system OEMs.

IT FITS SOME APPLICATIONS

Portable computer and automobile applications are Microsoft's primary targets. And while Microsoft also mentions some very embedded applications (e.g., temperature and robot controllers), this seems more like a pipe dream than reality.

The WinCE kernel is a multiprocess OS, and even if you develop for a single process, the application and kernel are separate processes, entailing much higher overhead on system calls and I/O accesses.

If your product is not cost sensitive, requires a graphical local display (especially one looking like Windows NT/95, as you see in Photo 1), or is sold as a portable computer, using WinCE as an embedded system makes sense. For the majority of embedded computing products, however, it's either overkill or it won't perform real-time tasks with any determinism.

I believe the best outcome of Microsoft's entry of Windows CE in the embedded space will be the tools developed for the

32-bit 'x86 market.

Visual Basic and Visual C++ support in multiple chip platforms, improvement of the Win32 API, and the promotion of a standard API (Win32 API) for embedded systems will help solidify that sector of the embedded computing market using 32-bit 'x86 platforms. **EPC**

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SOURCES

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Photo 1—This splash screen of a Windows CE system shows you the available applications.



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